



### Sub.: Non-Consumable Regenerateable Type Media Polishing System

THERMOPAC is an engineering and technology company specializing in the design, development, and implementation of advanced process technologies for **used oil re-refining and base oil finishing systems**.

The company provides complete **Engineering, Procurement, and Construction (EPC)** solutions for industrial processing plants, covering the full project lifecycle from process design and equipment fabrication to installation, automation integration, and plant commissioning.

Over the years, THERMOPAC has developed specialized technologies focused on upgrading recovered base oils into **commercially acceptable lubricant base stocks** with improved color, odor, and chemical stability.

Today, THERMOPAC technology is deployed across a global network of re-refining facilities. The company currently supports **more than 50 operating systems across five continents**, representing one of the largest installed bases of continuous base oil polishing technology in the re-refining sector.

Across these installations, **over 4,000 CPS polishing columns are currently in operation**, collectively processing **more than 1 million liters of refined base oil per day**. This extensive operating base provides THERMOPAC with a large body of practical operational experience and continuous performance feedback from real industrial plants.

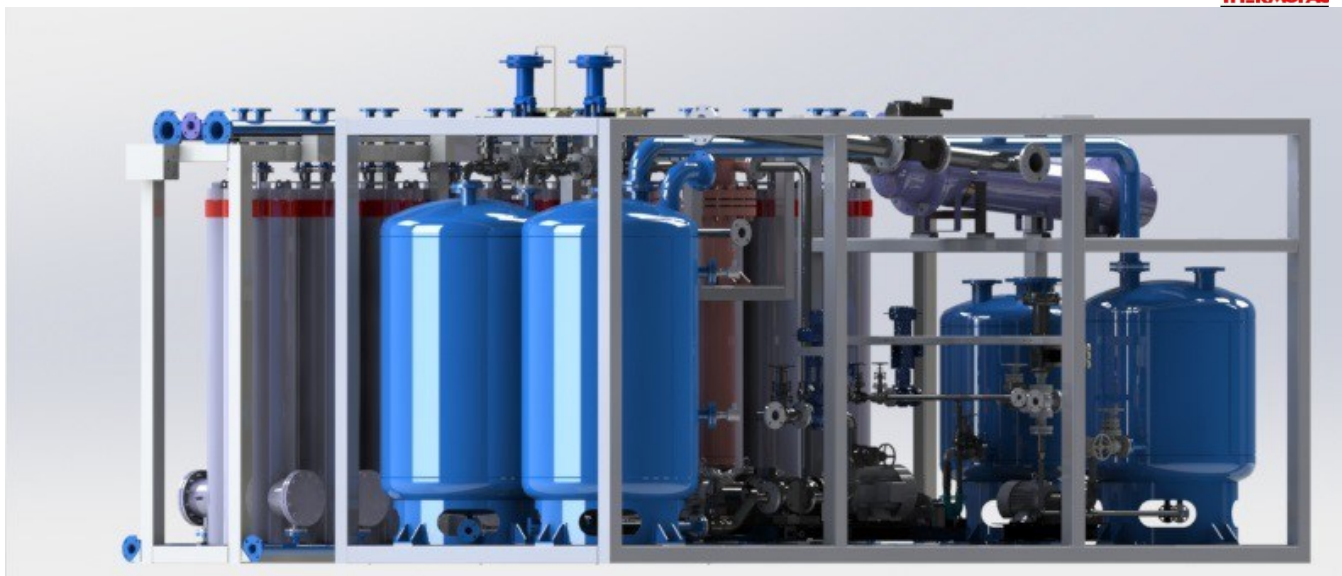
This global presence has enabled the company to refine its process technologies and deliver solutions that combine **high operational reliability, efficient regeneration cycles, and low operating costs**.

THERMOPAC's core capabilities include:

- Process technology development and engineering
- Design of continuous polishing systems for base oil finishing
- Equipment fabrication and modular process units
- Automation and PLC-SCADA control system integration
- Installation supervision and site support
- Plant commissioning and operator training

THERMOPAC systems are designed to integrate seamlessly with existing re-refining infrastructure, particularly downstream of **thin wiped film evaporators and vacuum distillation units**, where final polishing is required to achieve commercial base oil quality.

By combining proven adsorption technology, Regenerable polishing media, and integrated environmental protection systems, THERMOPAC delivers **cost-effective, reliable, and scalable solutions** tailored to the operational requirements of modern re-refining plants worldwide. The process and its advantages along with technical details are described in the following pages.



For **THERMOPAC**  
(Turnkey Engineering Solution Division)

Marketing Manager

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## THERMOPAC CONTINUOUS POLISHING SYSTEM (CPS)

### Engineering, Procurement & Construction (EPC) Proposal

Technology for Final Finishing of Recovered Base Oil from Thin Wiped Film Evaporator

Prepared by: **THERMOPAC Engineering Division**

Confidential – For Client Evaluation Only

#### Executive Summary

THERMOPAC proposes a **Continuous Polishing System (CPS)** designed to improve the quality of recovered base oil produced from a thin wiped film evaporator (TWFE) in used oil re-refining plants.

Although thin film evaporation effectively separates base oil fractions from contaminants and heavy residues, the recovered base oil may still contain trace impurities that affect its commercial quality. These impurities typically include residual sulfur compounds, color bodies formed during thermal degradation, oxidized hydrocarbons, and odor-causing molecules originating from degraded additives and sulfur species.

The Continuous Polishing System serves as the **final finishing stage** in the re-refining process and is designed to remove these remaining impurities, thereby improving the final product quality in terms of color, odor, and sulfur content.

The CPS technology uses **specialized Regenerable adsorption media packed in vertical columns**. As the recovered base oil passes through the packed bed, trace contaminants are selectively adsorbed by the media. Over time the adsorption media becomes saturated and must be regenerated to restore its adsorption capacity.

Unlike disposable filtration systems such as bleaching clay treatment, the CPS system uses **non-consumable Regenerable media**, allowing the polishing media to be reused through multiple adsorption–regeneration cycles. This significantly reduces operating costs and eliminates the need for continuous disposal of spent adsorbents.

The CPS system operates continuously through a multi-column configuration that allows simultaneous adsorption and regeneration operations. While one column is polishing the base oil stream, another column can undergo regeneration without interrupting plant operation.

During regeneration, the adsorbed contaminants are oxidized using controlled air injection at elevated temperature. The resulting exhaust gases contain hydrocarbons, sulfur compounds, and oxidation products that must be properly treated before release to the atmosphere.

To ensure environmental compliance, the CPS unit is integrated with a dedicated **exhaust gas treatment system**, consisting of gas cooling, cyclone separation, demisting, alkaline scrubbing, and thermal oxidation.

This integrated system ensures safe operation while meeting applicable emission standards.

The THERMOPAC CPS system provides the following key advantages:

- Continuous polishing operation
- Regenerable adsorption media with long service life
- Significant improvement in base oil color and odor
- Reduction of residual sulfur compounds
- Low operating and maintenance cost
- Integrated environmental emission control
- Fully automated PLC-SCADA control system

The CPS system is supplied as a **complete engineered package**, including process equipment, regeneration system, exhaust gas treatment system, instrumentation, and automation.

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### Project Understanding

In a typical used oil re-refining plant, the **thin wiped film evaporator (TWFE)** is used as the primary finishing stage to separate base oil fractions from heavy residues, degraded additives, and other high-boiling contaminants. The TWFE process efficiently recovers valuable lubricating base oil by operating under deep vacuum and controlled film evaporation conditions.

Although this process produces a high-quality recovered base oil with the desired boiling range, the product may still contain **trace contaminants that cannot be fully removed by distillation alone**. These residual compounds originate from thermal degradation of lubricants, additive decomposition, and long-term oxidation of used oils during service.

Typical residual contaminants remaining in the recovered base oil include:

- Sulfur compounds (thiophenes, mercaptans, sulfides)
- Oxidation products and degraded hydrocarbons
- Trace polar contaminants and nitrogen compounds
- Color bodies formed by polymerized oxidation products
- Odor-causing molecules derived from sulfur and nitrogen species

Even in small concentrations, these compounds can significantly affect the **appearance, odor, stability, and market acceptance** of the finished base oil.

For example:

#### Colour:

Recovered base oil may appear darker due to the presence of oxidized hydrocarbons and polymerized aromatic species formed during lubricant degradation. These compounds reduce the visual quality of the product and may limit its blending value in finished lubricants.

#### Odour:

Used lubricating oils often contain sulfur- and nitrogen-based molecules that generate strong unpleasant odors. Even after distillation, trace quantities of these compounds can remain in the recovered base oil, making the product unsuitable for commercial applications without further treatment.

#### Sulfur content:

Residual sulfur compounds that survive the distillation process can negatively affect oxidation stability and may limit the compatibility of the recovered base oil with modern additive packages. Reducing sulfur concentration is therefore essential to produce a stable and commercially acceptable base oil.

#### Product stability:

Trace oxidation products and polar contaminants can accelerate the aging of the base oil and contribute to color reversion during storage. These impurities can also affect the long-term stability of blended lubricants.

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Because these contaminants exist at very low concentrations and often have boiling points similar to base oil hydrocarbons, they cannot be effectively removed by further distillation or evaporation processes.

To address this challenge, a dedicated **Continuous Polishing System (CPS)** is installed downstream of the thin wiped film evaporator as the final finishing step in the re-refining process.

The CPS system is specifically designed to remove these residual impurities through **selective adsorption on Regenerable polishing media**. As the recovered base oil passes through the CPS columns, trace contaminants are captured by the adsorption media, resulting in significant improvements in the physical and chemical properties of the finished product.

The CPS polishing stage delivers the following key improvements:

**Colour improvement**

Adsorption media selectively removes oxidized hydrocarbons and color bodies, producing a significantly lighter and more visually appealing base oil.

**Odour reduction**

Sulfur- and nitrogen-based odor molecules are captured by the media, resulting in a neutral and commercially acceptable product odor.

**Sulfur reduction**

Residual sulfur compounds are adsorbed from the oil phase, reducing the sulfur content of the finished base oil.

**Improved product stability**

Removal of polar contaminants and oxidation products improves the long-term stability of the base oil and reduces the risk of color reversion during storage.

Through this polishing process, the CPS unit upgrades the recovered base oil into a **stable, clean, and commercially marketable lubricant base stock** suitable for blending into finished lubricant formulations.

The Continuous Polishing System therefore represents a **critical finishing stage in the used oil re-refining process**, ensuring that the recovered base oil meets the quality expectations of lubricant manufacturers and end users.

## Comparison of CPS Polishing Technology vs Hydroprocessing (Hydrotreating)

In the re-refining industry, several technologies are used to improve the quality of recovered base oil after distillation. Two of the most common approaches are **hydroprocessing (Hydrotreating)** and **adsorptive polishing using Continuous Polishing Systems (CPS)**.

Hydrotreating is widely used in petroleum refineries to remove sulfur, nitrogen, and aromatic compounds from hydrocarbon streams through catalytic reactions in the presence of hydrogen at high pressure and temperature. While this technology is highly effective, it requires complex equipment, high energy consumption, and significant capital investment.

The CPS technology developed by THERMOPAC provides an alternative finishing solution based on **selective adsorption and regenerative media technology**. This process operates at moderate conditions without hydrogen consumption and offers a cost-effective solution for improving the quality of recovered base oils produced in used oil re-refining plants.

The following table summarizes the key differences between the two approaches.

| Parameter               | CPS Continuous Polishing System                 | Hydroprocessing (Hydrotreating)            |
|-------------------------|-------------------------------------------------|--------------------------------------------|
| Process principle       | Adsorptive purification using Regenerable media | Catalytic hydrogenation reactions          |
| Operating pressure      | Near atmospheric                                | 80–150 bar                                 |
| Hydrogen requirement    | None                                            | Large hydrogen consumption                 |
| Operating temperature   | 60–80°C                                         | High (300–400°C reactor temperature)       |
| Reactor type            | Packed adsorption columns                       | High-pressure catalytic reactor            |
| CAPEX                   | Low to moderate                                 | Very high                                  |
| Operating cost          | Low                                             | High                                       |
| Utilities requirement   | Electricity, air, minor chemicals               | Hydrogen plant, high-pressure compressors  |
| Process complexity      | Simple                                          | Complex refinery process                   |
| Environmental footprint | Low                                             | High energy consumption                    |
| Regeneration            | Media regenerated in-situ                       | Catalyst replacement periodically required |
| Maintenance             | Simple                                          | Complex high-pressure equipment            |
| Installation footprint  | Compact modular system                          | Large refinery unit                        |

### Available Plant Sizes

THERMOPAC Continuous Polishing Systems are supplied as **modular process units** designed to integrate seamlessly with used oil re-refining plants operating downstream of thin wiped film evaporators. The CPS units are available in several standardized capacities to match the throughput of different plant sizes.

Each CPS system is engineered with multiple adsorption columns operating in continuous polishing mode, allowing simultaneous adsorption and regeneration to ensure uninterrupted plant operation.

The following standard CPS system capacities are available:

\*Actual throughput may vary depending on feed quality, sulfur content, and operating conditions, Processing time is ~ 36 to ~ 55 hours depend on single or double pass.

| Product Code    | System Model | Description                                            | Typical Processing Capacity* |
|-----------------|--------------|--------------------------------------------------------|------------------------------|
| CPS-BRA-CPS-60  | CPS 60       | Continuous Polishing System by Regenerative Adsorption | ~ 50000 LPD                  |
| CPS-BRA-CPS-120 | CPS 120      | Continuous Polishing System by Regenerative Adsorption | ~ 100000 LPD                 |
| CPS-BRA-CPS-180 | CPS 180      | Continuous Polishing System by Regenerative Adsorption | ~ 150000 LPD                 |
| CPS-BRA-CPS-200 | CPS 200      | Continuous Polishing System by Regenerative Adsorption | ~ 170000 LPD                 |
| CPS-BRA-CPS-240 | CPS 240      | Continuous Polishing System by Regenerative Adsorption | ~ 200000 LPD                 |

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## Modular System Design

The CPS polishing units are designed with a modular architecture that allows easy scaling of plant capacity. Larger re-refining facilities can install multiple CPS modules operating in parallel to handle higher base oil production rates.

Key features of the modular CPS design include:

- Standardized equipment configuration
- Simplified installation and integration
- Flexible capacity expansion
- Reduced engineering and fabrication time

This modular approach allows plant operators to select the CPS configuration that best matches the throughput of their existing thin wiped film evaporation system.

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## Continuous Operation Principle

Each CPS system uses a **multi-column configuration** that enables continuous polishing operation. While one or more columns operate in adsorption mode to treat the base oil stream, another column can undergo regeneration to restore the adsorption capacity of the media.

This configuration provides the following benefits:

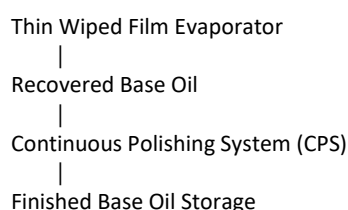
- Continuous polishing without process interruption
- Stable product quality
- Efficient regeneration cycles
- Optimized media utilization

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## System Integration

The CPS units are designed to integrate directly with the recovered base oil stream produced by the thin wiped film evaporator.

Typical integration point:



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## Custom Plant Capacities

In addition to the standard CPS models listed above, THERMOPAC can design custom polishing systems for larger re-refining facilities with higher production capacities.

Custom systems can be engineered to match specific plant requirements, including:

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- Higher throughput capacities
- Integration with existing plant infrastructure
- Customized regeneration cycles
- Site-specific environmental requirements

### CPS Polishing Technology

The Continuous Polishing System (CPS) developed by THERMOPAC utilizes **advanced adsorptive purification technology based on specially engineered Regenerable adsorption media**. The system is designed to perform the final upgrading stage for recovered base oils produced in used oil re-refining plants.

Following the primary separation processes such as vacuum distillation and thin wiped film evaporation, the recovered base oil generally meets the required boiling range specifications but may still contain trace impurities that negatively affect the product's commercial quality. These impurities often exist in very low concentrations and cannot be efficiently removed by further distillation because their physical properties are similar to those of the base oil hydrocarbons.

The CPS technology addresses this challenge through a **selective adsorption process**, where contaminants are removed from the oil phase by interaction with highly porous adsorption media contained within vertical packed columns.

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### Adsorptive Purification Principle

The CPS system operates on the principle of **adsorptive purification**, where undesirable molecules are selectively removed from the liquid phase by binding to the surface of the adsorption media.

The specially prepared polishing media used in the CPS system is designed to selectively capture and retain a range of contaminants typically present in recovered base oils, including:

- sulfur-containing molecules such as thiophenes, mercaptans, and sulfides
- oxidized hydrocarbons generated during lubricant degradation
- color bodies formed by polymerized aromatic compounds
- trace polar impurities and degraded additive residues

These molecules typically exhibit higher polarity and stronger surface affinity compared to the non-polar hydrocarbon molecules that make up the bulk of the base oil. As a result, the adsorption media selectively captures these contaminants while allowing the purified hydrocarbon molecules to pass through the packed bed.

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### Structure and Properties of the Adsorption Media

The CPS polishing media is specifically engineered to provide an optimal combination of **high adsorption capacity, mechanical strength, and regenerability**.

The media structure is characterized by:

- very high internal surface area
- interconnected microporous and mesoporous networks
- strong adsorption sites for polar molecules
- high thermal stability during regeneration

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The porous structure of the media creates a large internal surface area where contaminants can be captured. As the base oil flows through the packed bed, contaminant molecules diffuse into the pore network and are retained within the adsorbent structure through a combination of physical adsorption and surface interactions.

This porous architecture ensures efficient mass transfer and provides high adsorption capacity even at relatively low contaminant concentrations.

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### Selective Removal of Sulfur Compounds

Sulfur compounds present in recovered base oil are typically difficult to remove through distillation due to their boiling point similarity to hydrocarbon base oil molecules.

The CPS adsorption media exhibits strong affinity for sulfur-containing molecules, allowing selective removal of compounds such as:

- thiophenes
- benzothiophenes
- mercaptans
- sulfides

By capturing these sulfur compounds within the porous media structure, the CPS system significantly reduces the sulfur concentration in the finished base oil.

This sulfur reduction improves the oxidation stability of the oil and enhances compatibility with modern lubricant additive packages.

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### Removal of Colour Bodies

Recovered base oils often exhibit dark coloration due to the presence of oxidized hydrocarbons and polymerized aromatic compounds formed during the service life of the lubricant.

These compounds are typically larger and more polar than the base oil hydrocarbons and therefore exhibit strong affinity for the adsorption media.

As the oil passes through the CPS column, these color bodies are selectively captured within the media pores, resulting in a significant improvement in the visual appearance of the base oil.

The polishing process typically produces a **lighter and more visually stable product**, improving the commercial value of the recovered base oil.

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### Odor Reduction Mechanism

Odor in recovered base oils is commonly caused by sulfur compounds, nitrogen-containing molecules, and degraded additive fragments.

These molecules are generally polar and have strong interaction with the adsorption media. The CPS system effectively removes these odor-causing species during the polishing process.

As a result, the final base oil product exhibits **significantly reduced odor intensity**, making it suitable for blending into finished lubricants.

## Operating Conditions

The CPS polishing process operates under moderate conditions that are compatible with base oil finishing operations.

Typical operating parameters include:

- moderate temperatures suitable for base oil processing
- near-atmospheric operating pressure
- continuous liquid flow through packed adsorption columns

Unlike hydroprocessing technologies, the CPS system **does not require hydrogen, high-pressure reactors, or complex catalytic systems.**

This greatly simplifies the plant design and reduces both capital and operating costs.

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## Continuous Multi-Column Operation

To ensure uninterrupted polishing operation, the CPS system is typically configured with multiple adsorption columns operating in parallel.

While one or more columns operate in adsorption mode to treat the base oil stream, another column can undergo regeneration to restore its adsorption capacity.

This configuration provides several important advantages:

- continuous polishing operation without shutdown
- stable product quality
- optimized utilization of the adsorption media
- efficient regeneration cycles

The multi-column design ensures that the CPS system can operate continuously while maintaining consistent polishing performance.

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## Regenerable Media Concept

A key advantage of the CPS technology is the use of **non-consumable Regenerable adsorption media.**

Unlike disposable adsorption systems such as bleaching clay filtration, where the adsorbent must be replaced after saturation, the CPS media can be regenerated through controlled oxidation.

During regeneration, the adsorbed contaminants are removed from the media surface, restoring its adsorption capacity and allowing it to be reused for multiple cycles.

This Regenerable media concept significantly reduces operating costs and minimizes waste generation.

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## Exhaust Gas Treatment System

During the regeneration phase of the Continuous Polishing System (CPS), the adsorption media undergoes controlled oxidation in order to remove the contaminants accumulated during the adsorption cycle. This regeneration process generates a stream of off-gases composed of oxidation products and volatilized contaminants that must be carefully managed to ensure safe operation and environmental compliance.

The composition of the regeneration exhaust gas typically includes:

- Hydrocarbon vapors originating from degraded lubricating oil residues
- Carbon monoxide generated during partial oxidation reactions
- Carbon dioxide produced from complete oxidation reactions
- Water vapor generated during oxidation of hydrocarbons
- Hydrogen sulfide and sulfur compounds released from the adsorption media
- Light hydrocarbons and volatile organic compounds (VOCs)

In addition, trace quantities of aerosolized hydrocarbons and condensable vapors may be present in the gas stream due to the high temperature conditions within the regeneration column.

To manage these gases safely and to comply with environmental regulations, the CPS unit incorporates a **multi-stage exhaust gas treatment system** designed to progressively remove liquid contaminants, neutralize acid gases, and oxidize remaining hydrocarbons.

The exhaust gas treatment train typically consists of the following stages:

Knock-out separation  
Cyclone separator  
High-efficiency demister  
Alkaline scrubbing  
Thermal oxidation

Each stage performs a specific function in conditioning and treating the regeneration exhaust stream.

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### Knock-Out Separation

Following the gas cooling stage, the gas stream enters a **knock-out drum or separator vessel** designed to remove condensed liquid hydrocarbons and water from the gas phase.

The knock-out separator performs bulk liquid removal by allowing gravity separation of condensed droplets. The vessel is designed with sufficient residence time to allow heavier liquid droplets to settle and be collected at the bottom of the vessel.

Separated liquids may consist of:

- condensed hydrocarbons
- condensed water vapor
- oxidized oil residues

These liquids are periodically drained from the vessel and directed to appropriate waste or recovery handling systems.

By removing the bulk liquid content at this stage, the knock-out separator protects downstream equipment from liquid flooding and reduces the risk of fouling.

## Cyclone Separator

After the knock-out stage, the partially cleaned gas stream passes through a **cyclone separator**. The cyclone utilizes centrifugal forces to remove entrained liquid droplets and aerosol particles that may remain suspended in the gas stream.

Within the cyclone separator, the gas is introduced tangentially into a cylindrical chamber where it begins to rotate at high velocity. The centrifugal forces generated by this rotation cause heavier liquid droplets and particulate matter to migrate toward the outer walls of the cyclone.

These particles lose momentum and fall downward into a collection chamber where they are removed from the gas stream.

The cyclone separator is particularly effective for removing:

- hydrocarbon aerosols
- fine liquid droplets
- particulate contaminants

This stage significantly reduces the liquid loading entering the downstream gas treatment units.

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## High-Efficiency Demister

Although the cyclone separator removes a large portion of entrained droplets, very fine mist particles may still remain in the gas stream. To prevent carryover of these droplets into the scrubber and thermal oxidizer, the system incorporates a **high-efficiency demister**.

The demister typically consists of a structured mesh pad designed to capture fine droplets through inertial impaction and coalescence.

As the gas passes through the demister pad:

- small droplets collide with the mesh fibers
- droplets coalesce into larger droplets
- larger droplets fall back into the drainage system

The demister stage ensures that the gas entering the downstream equipment is essentially free of liquid carryover, thereby improving operational reliability and reducing maintenance requirements.

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## Alkaline Scrubbing

The next stage in the exhaust treatment system is the **alkaline scrubber**, which is designed to remove acid gases generated during the regeneration process.

The primary acid gases present in the regeneration exhaust stream include:

- hydrogen sulfide (H<sub>2</sub>S)
- sulfur oxides (SO<sub>2</sub>)

These compounds are removed through contact with an alkaline scrubbing solution, typically containing sodium hydroxide (NaOH).

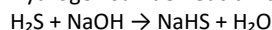
Inside the scrubber column, the gas flows through a packed bed where it comes into intimate contact with the circulating alkaline solution. Acid gases dissolve into the liquid phase and react with the alkaline solution to form neutral salts.

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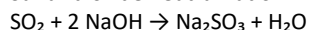
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Typical reactions include:

Hydrogen sulfide neutralization



Sulfur dioxide neutralization



The alkaline scrubber effectively reduces the concentration of sulfur compounds in the exhaust gas, preventing excessive SO<sub>x</sub> emissions in the final stack gas.

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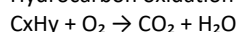
### Thermal Oxidizer

The final stage of the exhaust gas treatment system is the **thermal oxidizer**, which is responsible for the destruction of residual hydrocarbons and volatile organic compounds present in the gas stream.

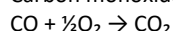
The thermal oxidizer operates at elevated temperatures, typically in the range of **750–850°C**, where organic compounds are oxidized to carbon dioxide and water.

Within the oxidizer chamber, the remaining hydrocarbons undergo complete combustion according to reactions such as:

Hydrocarbon oxidation



Carbon monoxide oxidation



The thermal oxidizer ensures that virtually all remaining volatile hydrocarbons are destroyed before the gas is discharged to the atmosphere.

The high-temperature oxidation process provides extremely high destruction efficiency for VOCs and ensures compliance with environmental emission standards.

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### Emission Compliance

The CPS exhaust gas treatment system is designed to ensure compliance with applicable environmental emission regulations governing industrial air emissions.

The integrated treatment process effectively removes liquid contaminants, neutralizes acid gases, and destroys volatile hydrocarbons generated during the regeneration process.

Typical emission targets for the treated exhaust gas include:

Sulfur dioxide (SO<sub>2</sub>) below regulatory emission limits

Total organic carbon (TOC) destruction through thermal oxidation

Carbon monoxide reduction through complete oxidation

The alkaline scrubbing stage significantly reduces the concentration of sulfur compounds entering the thermal oxidizer, thereby minimizing the formation of sulfur oxides in the final exhaust gas.

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Meanwhile, the thermal oxidizer ensures complete destruction of remaining hydrocarbons, resulting in very low levels of VOC emissions.

Through this multi-stage exhaust treatment approach, the CPS regeneration system achieves **safe and environmentally compliant operation**, meeting the air emission requirements typically applicable to industrial refining and chemical processing facilities.

The combination of condensation, separation, scrubbing, and thermal oxidation provides a robust and reliable system for managing regeneration exhaust gases while protecting both plant personnel and the environment.

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## Instrumentation & Control

The Continuous Polishing System (CPS) is equipped with a comprehensive instrumentation package designed to ensure **safe, reliable, and fully automated operation** of the polishing columns, regeneration system, and exhaust gas treatment system.

The instrumentation system enables real-time monitoring of all critical process parameters including temperature, pressure, flow, and gas composition. These measurements are continuously transmitted to the plant control system, allowing precise control of operating conditions and early detection of abnormal situations.

The following primary instrumentation is provided as part of the CPS system.

### Temperature Measurement

Temperature monitoring is critical for both the adsorption process and the regeneration cycle. Multiple temperature transmitters are installed at strategic locations throughout the system, including:

- CPS column inlet and outlet
- Packed bed temperature profile locations
- Regeneration air inlet
- Regeneration gas outlet
- Exhaust gas treatment system
- Thermal oxidizer interface

During regeneration, temperature sensors positioned at different elevations within the column monitor the movement of the **oxidation front through the adsorption media bed**. This ensures controlled regeneration and prevents overheating of the media.

All temperature transmitters are industrial-grade instruments designed for high-temperature service and are connected to the PLC control system for continuous monitoring and alarm generation.

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### Pressure Measurement

Pressure transmitters are installed throughout the system to monitor operating pressures and detect abnormal pressure conditions.

Typical pressure monitoring points include:

- CPS column inlet pressure
- packed bed differential pressure

## Control System Integration

All field instruments are connected to the plant PLC system through industrial communication networks. The measured process variables are continuously monitored and displayed on the operator interface, allowing plant personnel to maintain full visibility of the process conditions.

The instrumentation system supports:

- continuous process monitoring
- automatic process control
- alarm management
- safety interlock activation

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## Automation System

The CPS unit is supplied with a fully integrated **PLC–SCADA automation system** designed to ensure safe, reliable, and efficient plant operation.

The automation system performs real-time control of all critical process operations and provides a user-friendly interface for plant operators.

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## PLC-Based Control

A programmable logic controller (PLC) serves as the central control unit for the CPS system. The PLC executes control logic sequences that govern the operation of the adsorption columns, regeneration cycle, and exhaust gas treatment system.

The PLC system provides:

- real-time process control
- automatic sequencing of operations
- safety interlock management
- alarm and event logging

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## SCADA Operator Interface

A supervisory control and data acquisition (SCADA) system provides operators with graphical visualization of the entire CPS process.

The SCADA interface allows operators to:

- monitor process parameters in real time
- view equipment status and alarms
- control system operation
- review historical process trends

Graphical process displays provide a clear overview of system operation and facilitate rapid troubleshooting if required.

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## Automated Process Functions

The automation system controls all critical CPS operations, including:

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### **Column switching**

The PLC automatically switches between adsorption and regeneration columns to maintain continuous polishing operation without interrupting product flow.

### **Regeneration sequencing**

The regeneration cycle is fully automated and includes purge, heating, oxidation, and cooling phases.

### **Air flow control**

Regeneration air flow is precisely controlled to maintain optimal oxidation conditions and prevent excessive temperature rise.

### **Temperature monitoring**

Temperature profiles within the column are continuously monitored to ensure proper progression of the regeneration front.

### **Alarm management**

The automation system generates alarms and warnings when process parameters deviate from defined limits.

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## **Utilities Requirement**

The CPS system requires several standard plant utilities for normal operation.

### **Electrical Power**

Electric power is required for the operation of:

- feed pumps
- scrubber circulation pumps
- regeneration air blower
- instrumentation and control system

The electrical load depends on the specific plant capacity and equipment configuration.

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### **Cooling Water**

Cooling water is required for:

- gas cooling in the exhaust treatment system
- product cooling where required

Cooling water helps maintain stable operating temperatures and prevents overheating of downstream equipment.

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## Chemical Supply

An alkaline solution, typically sodium hydroxide (NaOH), is required for operation of the exhaust gas scrubber.

The scrubbing solution neutralizes acidic gases such as hydrogen sulfide and sulfur dioxide produced during regeneration.

The scrubber system includes a circulation tank and dosing system to maintain the required solution concentration.

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## Safety Systems

Safety is a critical design consideration in the CPS system. Multiple layers of safety protection are incorporated to ensure safe operation under all conditions.

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### LEL Monitoring

Hydrocarbon gas detectors continuously monitor the concentration of flammable gases within the exhaust system.

If hydrocarbon concentrations approach predefined limits, the control system automatically initiates protective actions.

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### Temperature Safety Interlocks

Temperature interlocks prevent overheating during regeneration. If excessive temperatures are detected within the adsorption column, the system automatically reduces regeneration air flow or shuts down the process.

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### Emergency Shutdown System

The CPS system includes an emergency shutdown (ESD) function that can immediately stop the process in the event of a critical fault.

Typical shutdown triggers include:

- excessive temperature
  - abnormal pressure conditions
  - high hydrocarbon concentration
  - equipment failure
- 

### Explosion Protection

Explosion protection measures are incorporated into the exhaust gas handling system to ensure safe operation.

These measures include:

- controlled air injection
- continuous gas monitoring
- safe operating procedures
- explosion-resistant equipment design

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## Installation & Commissioning

THERMOPAC provides comprehensive support for the installation and commissioning of the CPS system to ensure successful plant start-up.

Installation and commissioning services include:

- site installation supervision
  - equipment alignment and verification
  - piping and instrumentation checks
  - electrical and control system integration
  - cold and hot commissioning procedures
- 

## Start-Up Assistance

THERMOPAC process engineers assist with initial system start-up and optimization of operating conditions.

This includes:

- system flushing and preparation
  - initial adsorption operation
  - regeneration cycle commissioning
- 

## Operator Training

Operator training programs are conducted to ensure plant personnel fully understand system operation and maintenance requirements.

Training covers:

- system operation procedures
  - safety protocols
  - routine maintenance practices
- 

## Performance Testing

After commissioning, the CPS system undergoes performance testing to verify that all design specifications and operational targets are achieved.

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## Performance Guarantee

THERMOPAC guarantees that the CPS system will achieve the specified polishing performance when operated under the defined design conditions.

The performance guarantee typically includes:

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- reduction of color bodies
- removal of odor-causing compounds
- reduction of sulfur content
- stable continuous operation

Performance verification is conducted during the commissioning phase based on agreed testing procedures.

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## Exclusions

Unless specifically stated otherwise in the proposal, the following items are typically excluded from the CPS supply scope:

### Flow Measurement

Accurate flow control is essential for maintaining stable operating conditions in both the adsorption and regeneration processes.

Flow meters are installed to monitor:

- Base oil feed flow rate
- Finish oil flow rate
- Black oil flow rate

Flow measurement ensures proper mass balance control and allows the automation system to maintain optimal operating conditions.

### Gas Analysis

Gas analyzers are installed to monitor the composition of the regeneration exhaust gas stream. These analyzers provide important information about the oxidation process and ensure safe operation.

Typical gas analysis parameters include:

- oxygen concentration
- carbon monoxide (CO)
- hydrocarbon concentration (TOC)
- hydrogen sulfide (H<sub>2</sub>S)

Gas composition monitoring helps optimize the regeneration process and ensures that the exhaust treatment system operates within design limits.

### LEL Monitoring

Lower Explosive Limit (LEL) detectors are installed in the exhaust gas handling system to continuously monitor hydrocarbon concentrations in the gas stream.

These sensors provide early detection of potentially explosive gas mixtures and are integrated with the plant safety system. If hydrocarbon concentrations approach unsafe levels, automatic protective actions are initiated by the control system.

- civil works and foundations
- plant utility infrastructure
- external storage tanks and interconnecting piping
- upstream and downstream process units

These items may be supplied separately or by the client depending on project arrangements.

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## Spare Parts

THERMOPAC recommends maintaining a set of critical spare parts to ensure uninterrupted plant operation.

Typical spare parts include:

- pumps and pump seals
- control valves
- instrumentation components
- sensors and transmitters

A detailed spare parts list is provided as part of the project documentation.

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## After-Sales Support

THERMOPAC provides long-term technical support for all installed CPS systems.

After-sales services include:

- technical troubleshooting assistance
- spare parts supply
- process optimization support
- periodic performance evaluation

This ongoing support ensures reliable long-term operation of the CPS system.

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## Conclusion

The THERMOPAC Continuous Polishing System provides a highly reliable and cost-effective solution for upgrading recovered base oil produced in used oil re-refining plants.

By utilizing Regenerable adsorption media, the CPS system effectively removes residual impurities responsible for color degradation, unpleasant odor, and elevated sulfur content.

The multi-column continuous operation design ensures stable polishing performance while the integrated regeneration and exhaust treatment systems enable efficient media regeneration and environmentally compliant operation.

Through the combination of proven adsorption technology, advanced automation, and robust environmental protection systems, the CPS unit delivers consistent product quality and long-term operational reliability.

The THERMOPAC CPS system therefore represents a **complete and integrated finishing solution for modern base oil re-refining facilities worldwide.**

**(Turnkey Engineering Solution Division)**

Marketing Manager

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